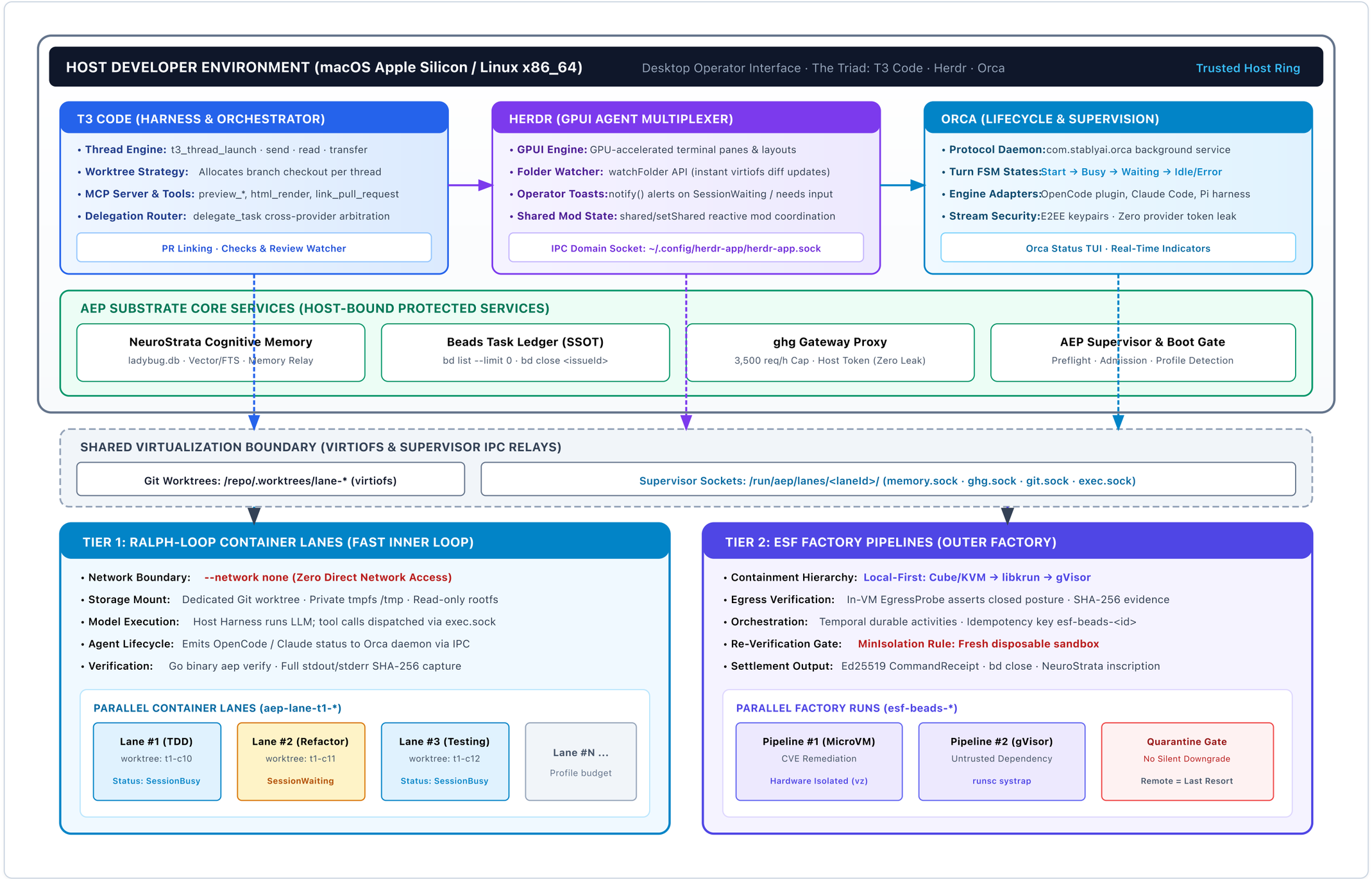
Integration Patterns Across T3 Code, Herdr, and Orca

System Integration Architecture · Version: v1.2 (Immutable Artifact) · **Date:** 2026-11-09 · **Branch:** `docs/dual-lane-governance-proposal` · **PR:** #133 · **Companion Docs:** `aep-architecture-diagrams-v1.2.html`, `aep-hld-v1.2.html`, `aep-lld-v1.2.html`

This reference artifact formalizes the integration topology connecting the developer desktop environment to the Autonomous Engineering Platform (AEP) substrate. It establishes the technical contracts, event pipelines, and credential boundaries between **T3 Code** (AI-assisted harness and thread orchestrator), **Herdr** (GPUI agent multiplexer and terminal harness), and **Orca** (agent lifecycle, telemetry, and supervision daemon).

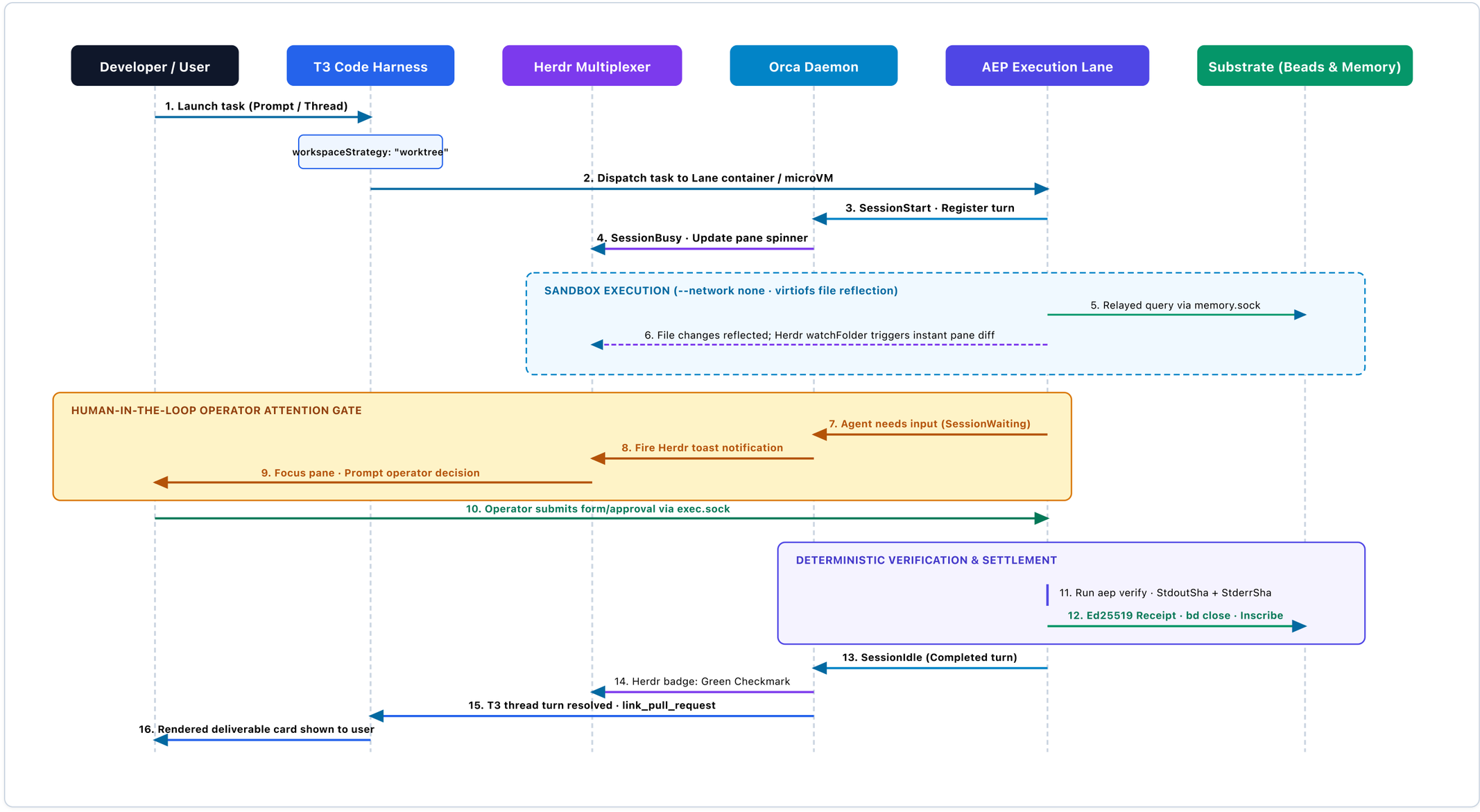
1. System Integration Topology: T3 Code, Herdr, Orca & AEP Substrate

Full architectural mapping showing desktop operator tools, shared virtiofs filesystems, supervisor domain socket relays, and dual-lane execution containment.



2. Interaction Sequence & Agent Lifecycle State Machine

Sequential flow demonstrating task allocation, instant file reflection via virtiofs, Orca session lifecycle states, human-in-the-loop attention routing, and verifiable settlement.



3. Detailed Specifications of the Four Integration Patterns

Pattern A: Workspace Reflection (Virtiofs)

T3 Code provisions Git worktrees bound to isolated thread IDs. Virtiofs reflects all file modifications between container lanes and the host instantly. Herdr's `watchFolder` receives kernel events without polling, instantly refreshing syntax-highlighted diffs in open multiplexer panes.

Pattern B: Agent Lifecycle FSM (Orca)

OpenCode, Claude Code, and Pi harness execution states flow into Orca's daemon over local domain sockets. State transitions (`SessionStart` → `SessionBusy` → `SessionWaiting` → `SessionIdle`) drive Herdr's attention routing and T3 Code's turn completions.

Pattern C: Zero-Token-Leak Socket Relays

Execution lanes are locked down with `--network none` or measured `EgressProbe` boundaries. GitHub PATs and LLM provider secrets stay strictly on the host. Sockets mounted at `/run/aep/lanes/<laneId>/` proxy memory queries, GitHub calls (capped at 3,500 req/h), and model harness interactions.

Pattern D: Cryptographic Settlement

Verification executes via the Go static binary `aep verify`, hashing every stdout/stderr byte. RFC 8785 canonical receipts signed with Ed25519 are mandatory for `bd close <issueId>` ledger settlement, continuous memory inscription, and automatic GitHub PR linking.

4. Raw Mermaid Diagrams Source (v1.2)

The exact Mermaid source definitions used for this release, compliant with standard Mermaid parsers:

4.1 Integration Topology (Flowchart TD)

```
flowchart TD
  subgraph HOST_DESKTOP["Host Developer Environment (macOS / Linux)"]
    subgraph T3_CODE["T3 Code (Harness & Orchestrator)"]
      T3_Thread["Thread Engine\n(Launch / Send / Read / Transfer)"]
      T3_Worktree["Worktree Strategy\n(Branch Isolation)"]
      T3_MCP["T3 MCP Server\n(Previews, Devices, PR Link)"]
      T3_Delegator["Delegate Task\n(Cross-Provider Routing)"]
    end

    subgraph HERDR["Herdr (GPUI Agent Multiplexer)"]
      Herdr_Core["Herdr Core Daemon\n(herdr.d.ts / IPC Socket)"]
      Herdr_UI["GPUI Terminal Panes\n(Multi-Agent Multiplexer)"]
      Herdr_Watcher["Folder Watcher\n(watchFolder API)"]
      Herdr_Toasts["Notification Toasts\n(User Attention Alerts)"]
    end

    subgraph ORCA["Orca (Agent Lifecycle & Supervision)"]
      Orca_Daemon["Orca Protocol Daemon\n(com.stablyai.orca)"]
      Orca_FSM["Session Lifecycle FSM\n(Start -> Busy -> Waiting -> Idle)"]
      Orca_TUI["Orca Status TUI\n(Real-Time Indicators)"]
      Orca_E2EE["Telemetry & E2EE Keys\n(Zero Token Leakage)"]
    end

    subgraph SUBSTRATE["AEP Substrate Core Services"]
      NS_Memory["NeuroStrata Cognitive Memory\n(ladybug.db / Vector / FTS)"]
      Beads_Ledger["Beads Issue Ledger\n(Single Source of Truth)"]
      GHG_Gateway["GHG GitHub Gateway\n(3,500 req/h Cap / Host Token)"]
      AEP_Supervisor["AEP Supervisor & Boot Gate\n(Preflight & Admission Control)"]
    end
  end

  subgraph VIRT_FILESYSTEM["Shared Filesystem & Relays (Virtiofs)"]
    Git_Worktrees["Git Worktree Storage\n(/repo/.worktrees/lane-*)"]
    Unix_Sockets["Supervisor Socket Relays\n(/run/aep/lanes/<laneId>/)"]
  end

  subgraph EXECUTION_TIERS["Dual-Lane Execution Substrate"]
    subgraph TIER1["Tier 1: Ralph-Loop Lanes (Fast Inner Loop)"]
      Ralph_Container["OCI Container (--network none)\nPrivate tmpfs / Read-Only rootfs"]
      Ralph_Agent["Agent Loop (TDD / Plan / Verify)\nHost Model via exec.sock"]
      Ralph_Verifier["aep verify Binary\nFull-stream Stdout/Stderr SHA-256"]
    end

    subgraph TIER2["Tier 2: ESF Factory Pipelines (Outer Factory)"]
      ESF_Temporal["Temporal Durable Workflow\n(esf-beads-<id>)"]
      ESF_Sandbox["Local-First Sandbox\n(Cube/KVM -> libkrun -> gVisor)"]
      ESF_Probe["In-VM EgressProbe\n(Measured Closed Boundary)"]
      ESF_Reverifier["Fresh Disposable Sandbox\n(MinIsolation Rule)"]
    end
  end

  T3_Worktree -->|"1. Allocates isolated branch"| Git_Worktrees
  Git_Worktrees -->|"2. Virtiofs event reflection"| Herdr_Watcher
  Herdr_Watcher -->|"3. Instant UI diff refresh"| Herdr_UI

  T3_Thread -->|"Spawns lane task"| AEP_Supervisor
  AEP_Supervisor -->|"Admits & provisions"| Ralph_Container
```

```
  AEP_Supervisor -->|"Dispatches factory task"| ESF_Temporal

  Ralph_Agent -->|"Lifecycle events"| Orca_FSM
  ESF_Temporal -->|"Pipeline status"| Orca_FSM
  Orca_FSM -->|"Broadcasts Waiting / Idle"| Orca_Daemon
  Orca_Daemon -->|"IPC socket relay"| Herdr_Core
  Herdr_Core -->|"Fires desktop toast"| Herdr_Toasts
  Herdr_Core -->|"Updates pane status badge"| Herdr_UI
  Orca_Daemon -->|"Syncs thread state"| T3_Thread

  Ralph_Container -->|"Zero direct network"| Unix_Sockets
  ESF_Sandbox -->|"Measured egress proxy"| Unix_Sockets
  Unix_Sockets -->|"Proxy GitHub calls (no token in lane)"| GHG_Gateway
  Unix_Sockets -->|"Memory retrieval / inscription"| NS_Memory

  Ralph_Verifier -->|"Ed25519 CommandReceipt"| Beads_Ledger
  ESF_Reverifier -->|"Verified patch digest"| Beads_Ledger
  Beads_Ledger -->|"Atomic settlement (bd close)"| NS_Memory
  Beads_Ledger -->|"Settled state trigger"| T3_MCP
  T3_MCP -->|"link_pull_request / PR review"| T3_Thread
```

4.2 Interaction Lifecycle (Sequence Diagram)

```
sequenceDiagram
  autonumber
  actor User as Developer / Operator
  participant T3 as T3 Code
  participant Herdr as Herdr Multiplexer
  participant Orca as Orca Daemon
  participant AEP as AEP Supervisor & Lanes
  participant Substrate as Substrate (Beads & Memory)

  User->>T3: Launch engineering task (Thread / Goal)
  T3->>T3: Allocate Git worktree (workspaceStrategy: "worktree")
  T3->>AEP: Dispatch task to Lane (Tier 1 Ralph or Tier 2 ESF)
  AEP->>Orca: Agent session started (SessionStart)
  Orca->>Herdr: Post status: Lane Busy (SessionBusy)
  Herdr->>Herdr: Render active spinner in agent pane

  rect rgb(240, 248, 255)
    note over AEP,Substrate: Execution inside --network none sandbox
    AEP->>AEP: Run agent loop & execute code changes
    AEP->>Substrate: Relayed tool/memory query via supervisor socket
    AEP->>T3: File writes reflected via virtiofs
    Herdr->>Herdr: watchFolder detects changes (instant UI reflection)
  end

  alt Agent requires operator input / approval
    AEP->>Orca: Form / permission requested
    Orca->>Orca: Transition FSM to SessionWaiting
    Orca->>Herdr: Broadcast waiting state over IPC socket
    Herdr->>User: Fire notification toast & focus agent pane
    User->>Herdr: Review diff & submit approval
    Herdr->>AEP: Pass input back via exec.sock
  end

  rect rgb(245, 243, 255)
    note over AEP,Substrate: Deterministic Verification & Settlement
    AEP->>AEP: Run aep verify (full StdoutSha / StderrSha)
    AEP->>AEP: Sign CommandReceipt with Ed25519
    AEP->>Substrate: Settle issue (bd close <id>) & inscribe NeuroStrata
  end
```

```
    AEP->>Orca: Session completed (SessionIdle)
    Orca->>Herdr: Update terminal badge to green checkmark
    Orca->>T3: Report thread turn completion
    T3->>T3: Call link_pull_request with verified branch PR
    T3->>User: Display completed task summary & rendered deliverables
```
